

Chapter 5

Mobile Radio Propagation: Small-Scale Fading and Multipath

Presented By~

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Small Scale Fading

- Small-scale fading, or simply fading, is caused by interference between two or more versions of the transmitted signal which arrive at the receiver at slightly different times.
- These waves, called multipath waves, combine at the receiver antenna to give a resultant signal which can vary widely in amplitude and phase, depending on the distribution of the intensity and relative propagation time of the waves and the bandwidth of the transmitted signal.

Small-Scale Multipath Propagation

- Multipath in the radio channel creates small-scale fading effects. The three most important effects are:
- Rapid changes in signal strength over a small travel distance or time interval
- Random frequency modulation due to varying Doppler shifts on different multipath signals
- Time dispersion (echoes) caused by multipath propagation delays.

Factors Influencing Small-Scale Fading

- Multipath propagation
- Speed of the mobile
- Speed of surrounding objects
- The transmission bandwidth of the signal

Doppler Shift

- Due to the relative motion between the mobile and the base station, each multipath wave experiences an apparent shift in frequency.
- The shift in received signal frequency due to motion is called the Doppler shift, and is directly proportional to the velocity and direction of motion of the mobile with respect to the direction of arrival of the received multipath wave.

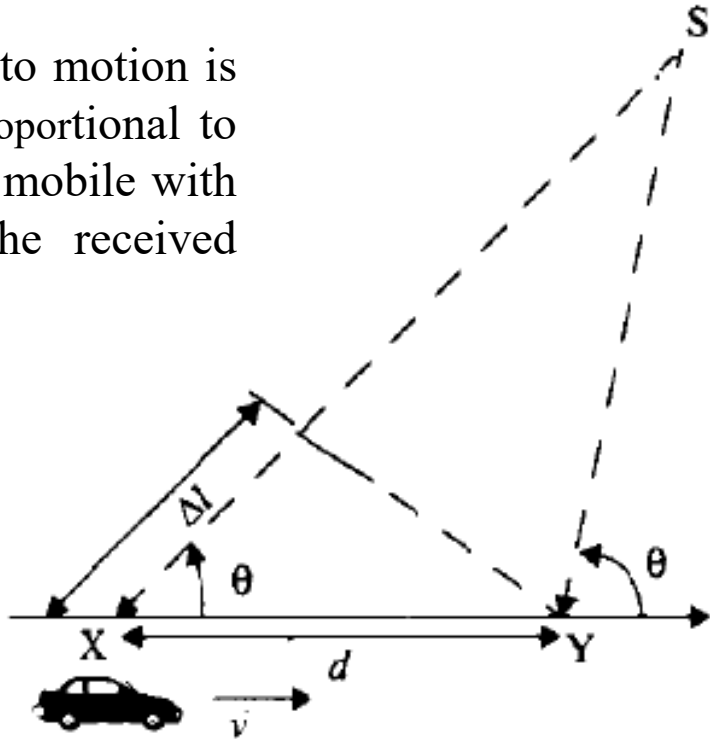


Figure 4.1
Illustration of Doppler effect.

Doppler Shift

Consider a mobile moving at a constant velocity v , along a path segment having length d between points X and Y, while it receives signals from a remote source S, as illustrated in Figure 4.1. The difference in path lengths traveled by the wave from source S to the mobile at points X and Y is $\Delta l = d \cos \theta = v \Delta t \cos \theta$, where Δt is the time required for the mobile to travel from X to Y, and θ is assumed to be the same at points X and Y since the source is assumed to be very far away. The phase change in the received signal due to the difference in path lengths is therefore

$$\Delta \phi = \frac{2\pi \Delta l}{\lambda} = \frac{2\pi v \Delta t}{\lambda} \cos \theta \quad (4.1)$$

and hence the apparent change in frequency, or Doppler shift, is given by f_d , where

$$f_d = \frac{1}{2\pi} \cdot \frac{\Delta \phi}{\Delta t} = \frac{v}{\lambda} \cdot \cos \theta \quad (4.2)$$

Doppler Shift

- Equation (4.2) relates the Doppler shift to the mobile velocity and the spatial angle between the direction of motion of the mobile and the direction of arrival of the wave.
- It can be seen from equation (4.2) that if the mobile is moving toward the direction of arrival of the wave, the Doppler shift is positive (i.e., the apparent received frequency is increased), and if the mobile is moving away from the direction of arrival of the wave, the Doppler shift is negative.
- Multipath components from a CW signal which arrive from different directions contribute to Doppler spreading of the received signal, thus increasing the signal bandwidth.

Doppler Shift

Example 4.1

Consider a transmitter which radiates a sinusoidal carrier frequency of 1850 MHz. For a vehicle moving 60 mph, compute the received carrier frequency if the mobile is moving (a) directly towards the transmitter, (b) directly away from the transmitter, (c) in a direction which is perpendicular to the direction of arrival of the transmitted signal.

Solution to Example 4.1

Given:

Carrier frequency $f_c = 1850 \text{ MHz}$

Therefore, wavelength $\lambda = c/f_c = \frac{3 \times 10^8}{1850 \times 10^6} = 0.162 \text{ m}$

Vehicle speed $v = 60 \text{ mph} = 26.82 \text{ m/s}$

(a) The vehicle is moving directly towards the transmitter.

The Doppler shift in this case is positive and the received frequency is given by equation (4.2)

$$f = f_c + f_d = 1850 \times 10^6 + \frac{26.82}{0.162} = 1850.00016 \text{ MHz}$$

(b) The vehicle is moving directly away from the transmitter.

The Doppler shift in this case is negative and hence the received frequency is given by

Doppler Shift

$$f = f_c - f_d = 1850 \times 10^6 - \frac{26.82}{0.162} = 1849.999834 \text{ MHz}$$

(c) The vehicle is moving perpendicular to the angle of arrival of the transmitted signal.

In this case, $\theta = 90^\circ$, $\cos\theta = 0$, and there is no Doppler shift.

The received signal frequency is the same as the transmitted frequency of 1850 MHz.

Doppler Shift

Example 5.1

In the U.S. digital cellular system, if $f_c = 900$ MHz and the mobile velocity is 70 km/hr, calculate the received carrier frequency if the mobile (a) directly toward the transmitter (Positive Doppler Shift), (b) directly away from the transmitter (Negative Doppler Shift), and (c) in a direction perpendicular to the direction of the arrival of the transmitted signal.

Doppler Shift

Example 5.1

In the U.S. digital cellular system, if $f_c = 900$ MHz and the mobile velocity is 70 km/hr, calculate the received carrier frequency if the mobile (a) directly toward the transmitter (Positive Doppler Shift), (b) directly away from the transmitter (Negative Doppler Shift), and (c) in a direction perpendicular to the direction of the arrival of the transmitted signal.

Solution

Given:

Carrier frequency $f_c = 900$ MHz

$$\text{Therefore, wavelength } \lambda = c/f_c = \frac{3 \times 10^8}{900 \times 10^6} = 1/3 \text{ m} = 0.33 \text{ m}$$

Vehicle speed, $v = 70 \times 1000/60 \times 60 = 19.44$ m/s

(a) The vehicle is moving directly toward the transmitter.

The received frequency is

$$f = f_c + f_d = 900 \times 10^6 + \frac{19.44}{0.33} = 900.0000589 \text{ MHz}$$

(b) The vehicle is moving directly away from the transmitter.

The received frequency is given by

$$f = f_c - f_d = 900 \times 10^6 - \frac{19.44}{0.33} = 899.9999411 \text{ MHz}$$

(c) The vehicle is moving perpendicular to the angle of arrival of the transmitted signal.

In this case, $\theta = 90^\circ$, $\cos \theta = 0$, and there is no Doppler shift.

The received signal frequency is the same as the transmitted frequency of 900 MHz.

Impulse Response Model of a Multipath Channel

- The small-scale variations of a mobile radio signal can be directly related to the impulse response of the mobile radio channel. The impulse response is a wideband channel characterization and contains all information necessary to simulate or analyze any type of radio transmission through the channel.
- This stems from the fact that a mobile radio channel may be modeled as a linear filter with a time varying impulse response, where the time variation is due to receiver motion in space.
- The filtering nature of the channel is caused by the summation of amplitudes and delays of the multiple arriving waves at any instant of time.
- The impulse response is a useful characterization of the channel, since it may be used to predict and compare the performance of many different mobile communication systems and transmission bandwidths for a particular mobile channel condition.

Impulse Response Model of a Multipath Channel

- To show that a mobile radio channel may be modeled as a linear filter with a time varying impulse response, consider the case where time variation is due strictly to receiver motion in space.



Figure 4.2

The mobile radio channel as a function of time and space.

- The receiver moves along the ground at some constant velocity v . For a fixed position d , the channel between the transmitter and the receiver can be modeled as a linear time invariant system.
- However, due to the different multipath waves which have propagation delays which vary over different spatial locations of the receiver, the impulse response of the linear time invariant channel should be a function of the position of the receiver. That is, the channel impulse response can be expressed as $h(d,t)$.
- Let $x(t)$ represent the transmitted signal, then the received signal $y(d,t)$ at position d can be expressed as a convolution of $x(t)$ with $h(d,t)$.

Impulse Response Model of a Multipath Channel

$$y(d, t) = x(t) \otimes h(d, t) = \int_{-\infty}^{\infty} x(\tau) h(d, t - \tau) d\tau \quad (4.3)$$

For a causal system, $h(d, t) = 0$ for $t < 0$, thus equation (4.3) reduces to

$$y(d, t) = \int_{-\infty}^t x(\tau) h(d, t - \tau) d\tau \quad (4.4)$$

Since the receiver moves along the ground at a constant velocity v , the position of the receiver can be expressed as

$$d = vt \quad (4.5)$$

Substituting (4.5) in (4.4), we obtain

$$y(vt, t) = \int_{-\infty}^t x(\tau) h(vt, t - \tau) d\tau \quad (4.6)$$

Since v is a constant, $y(vt, t)$ is just a function of t . Therefore, equation (4.6) can be expressed as

$$y(t) = \int_{-\infty}^t x(\tau) h(vt, t - \tau) d\tau = x(t) \otimes h(vt, t) = x(t) \otimes h(d, t) \quad (4.7)$$

From equation (4.7) it is clear that the mobile radio channel can be modeled as a linear time varying channel, where the channel changes with time and distance.

Impulse Response Model of a Multipath Channel

Since v may be assumed constant over a short time (or distance) interval, we may let $x(t)$ represent the transmitted bandpass waveform, $y(t)$ the received waveform, and $h(t, \tau)$ the impulse response of the time varying multipath radio channel. The impulse response $h(t, \tau)$ completely characterizes the channel and is a function of both t and τ . The variable t represents the time variations due to motion, whereas τ represents the channel multipath delay for a fixed value of t . One may think of τ as being a vernier adjustment of time. The received signal $y(t)$ can be expressed as a convolution of the transmitted signal $x(t)$ with the channel impulse response (see Figure 4.3a).

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t, \tau)d\tau = x(t) \otimes h(t, \tau) \quad (4.8)$$

If the multipath channel is assumed to be a bandlimited bandpass channel, which is reasonable, then $h(t, \tau)$ may be equivalently described by a complex baseband impulse response $h_b(t, \tau)$ with the input and output being the complex envelope representations of the transmitted and received signals, respectively (see Figure 4.3b). That is,

$$r(t) = c(t) \otimes \frac{1}{2}h_b(t, \tau) \quad (4.9)$$

Impulse Response Model of a Multipath Channel

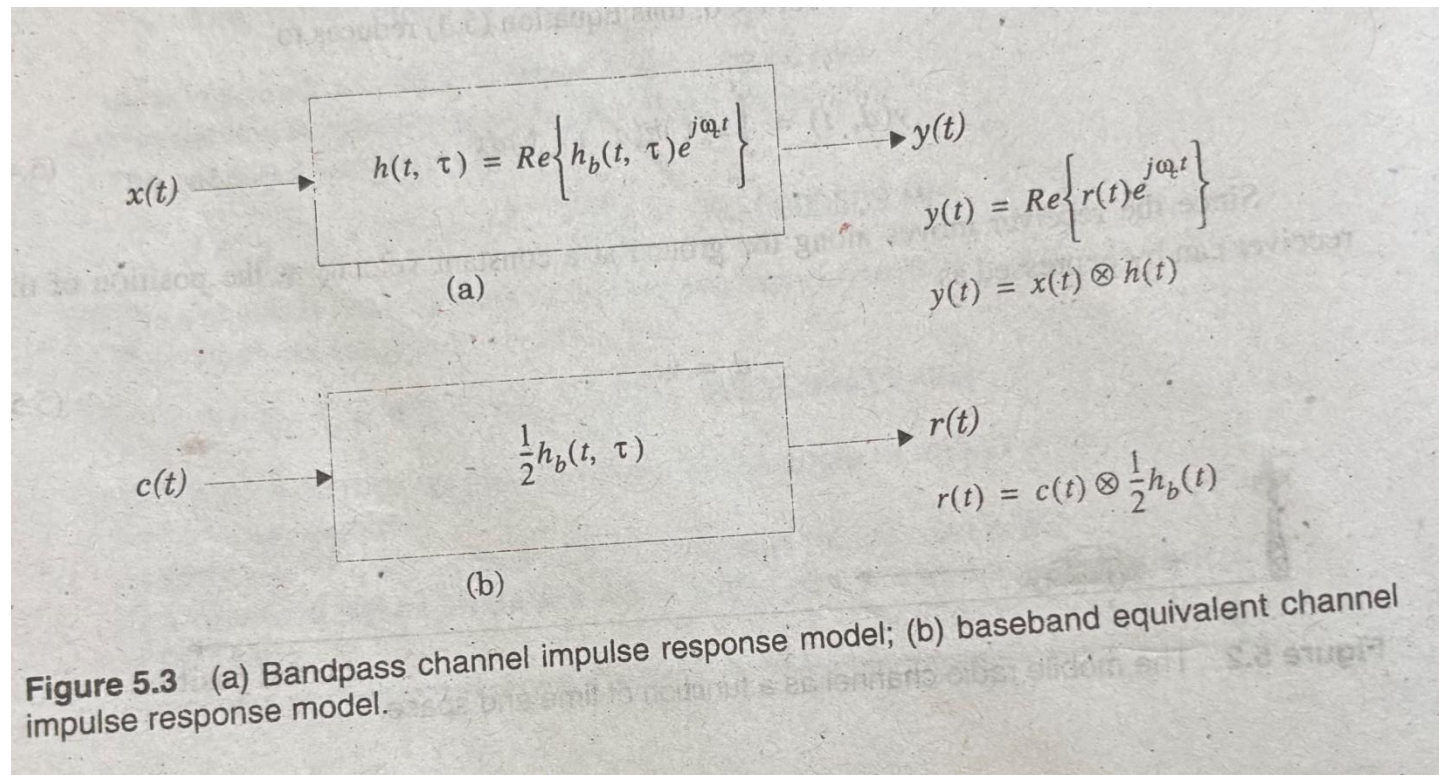
where $c(t)$ and $r(t)$ are the complex envelopes of $x(t)$ and $y(t)$, defined as

$$x(t) = \text{Re}\{c(t)\exp(j2\pi f_c t)\} \quad (4.10)$$

$$y(t) = \text{Re}\{r(t)\exp(j2\pi f_c t)\} \quad (4.11)$$

The factor of $1/2$ in equation (4.9) is due to the properties of the complex envelope, in order to represent the passband radio system at baseband. The low-pass characterization removes the high frequency variations caused by the carrier, making the signal analytically easier to handle. It is shown by Couch [Cou93] that the average power of a bandpass signal $\overline{x^2(t)}$ is equal to $\frac{1}{2}\overline{|c(t)|^2}$, where the overbar denotes ensemble average for a stochastic signal, or time average for a deterministic or ergodic stochastic signal.

Impulse Response Model of a Multipath Channel



Impulse Response Model of a Multipath Channel

It is useful to discretize the multipath delay axis τ of the impulse response into equal time delay segments called *excess delay bins*, where each bin has a time delay width equal to $\tau_{i+1} - \tau_i$, where τ_0 is equal to 0, and represents the first arriving signal at the receiver. Letting $i = 0$, it is seen that $\tau_1 - \tau_0$ is equal to the time delay bin width given by $\Delta\tau$. For convention, $\tau_0 = 0$, $\tau_1 = \Delta\tau$, and $\tau_i = i\Delta\tau$, for $i = 0$ to $N - 1$, where N represents the total number of possible equally-spaced multipath components, including the first arriving component. Any number of multipath signals received within the i th bin are represented by a single resolvable multipath component having delay τ_i . This technique of quantizing the delay bins determines the time delay resolution of the channel model, and the useful frequency span of the model can be shown to be $1/(2\Delta\tau)$. That is, the model may be used to analyze transmitted signals having bandwidths which are less than $1/(2\Delta\tau)$. Note that $\tau_0 = 0$ is the excess time delay

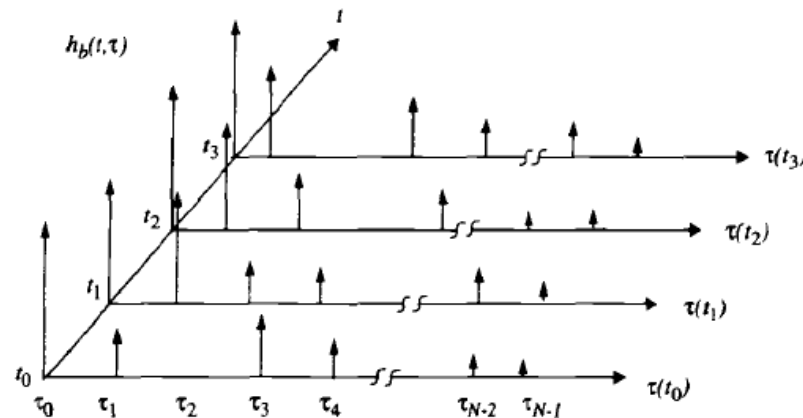


Figure 4.4

An example of the time varying discrete-time impulse response model for a multipath radio channel.

Impulse Response Model of a Multipath Channel

of the first arriving multipath component, and neglects the propagation delay between the transmitter and receiver. *Excess delay* is the relative delay of the i th multipath component as compared to the first arriving component and is given by τ_i . The *maximum excess delay* of the channel is given by $N\Delta\tau$.

Since the received signal in a multipath channel consists of a series of attenuated, time-delayed, phase shifted replicas of the transmitted signal, the baseband impulse response of a multipath channel can be expressed as

$$h_b(t, \tau) = \sum_{i=0}^{N-1} a_i(t, \tau) \exp[j(2\pi f_c \tau_i(t) + \phi_i(t, \tau))] \delta(\tau - \tau_i(t)) \quad (4.12)$$

where $a_i(t, \tau)$ and $\tau_i(t)$ are the real amplitudes and excess delays, respectively, of i th multipath component at time t [Tur72]. The phase term $2\pi f_c \tau_i(t) + \phi_i(t, \tau)$ in (4.12) represents the phase shift due to free space propagation of the i th multipath component, plus any additional phase shifts which are encountered in the channel. In general, the phase term is simply represented by

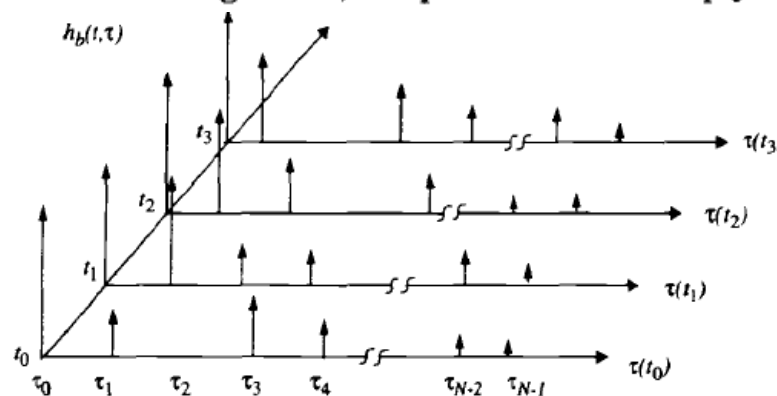


Figure 4.4

An example of the time varying discrete-time impulse response model for a multipath radio channel.

Impulse Response Model of a Multipath Channel

If the channel impulse response is assumed to be time invariant, or is at least wide sense stationary over a small-scale time or distance interval, then the channel impulse response may be simplified as

$$h_b(\tau) = \sum_{i=0}^{N-1} a_i \exp(-j\theta_i) \delta(\tau - \tau_i) \quad (4.13)$$

When measuring or predicting $h_b(\tau)$, a probing pulse $p(t)$ which approximates a delta function is used at the transmitter. That is

$$p(t) \approx \delta(t - \tau) \quad (4.14)$$

is used to sound the channel to determine $h_b(\tau)$.

For small-scale channel modeling, the *power delay profile* of the channel is found by taking the spatial average of $|h_b(t; \tau)|^2$ over a local area. By making several local area measurements of $|h_b(t; \tau)|^2$ in different locations, it is possible to build an ensemble of power delay profiles, each one representing a possible small-scale multipath channel state [Rap91a].

Based on work by Cox [Cox72], [Cox75], if $p(t)$ has a time duration much smaller than the impulse response of the multipath channel, $p(t)$ does not need to be deconvolved from the received signal $r(t)$ in order to determine relative multipath signal strengths. The received power delay profile in a local area is given by

$$P(t; \tau) \approx k |h_b(t; \tau)|^2 \quad (4.15)$$

and many snapshots of $|h_b(t; \tau)|^2$ are typically averaged over a local (small-scale) area to provide a single time-invariant multipath power delay profile $P(\tau)$. The gain k in equation (4.15) relates the transmitted power in the probing pulse $p(t)$ to the total power received in a multipath delay profile.

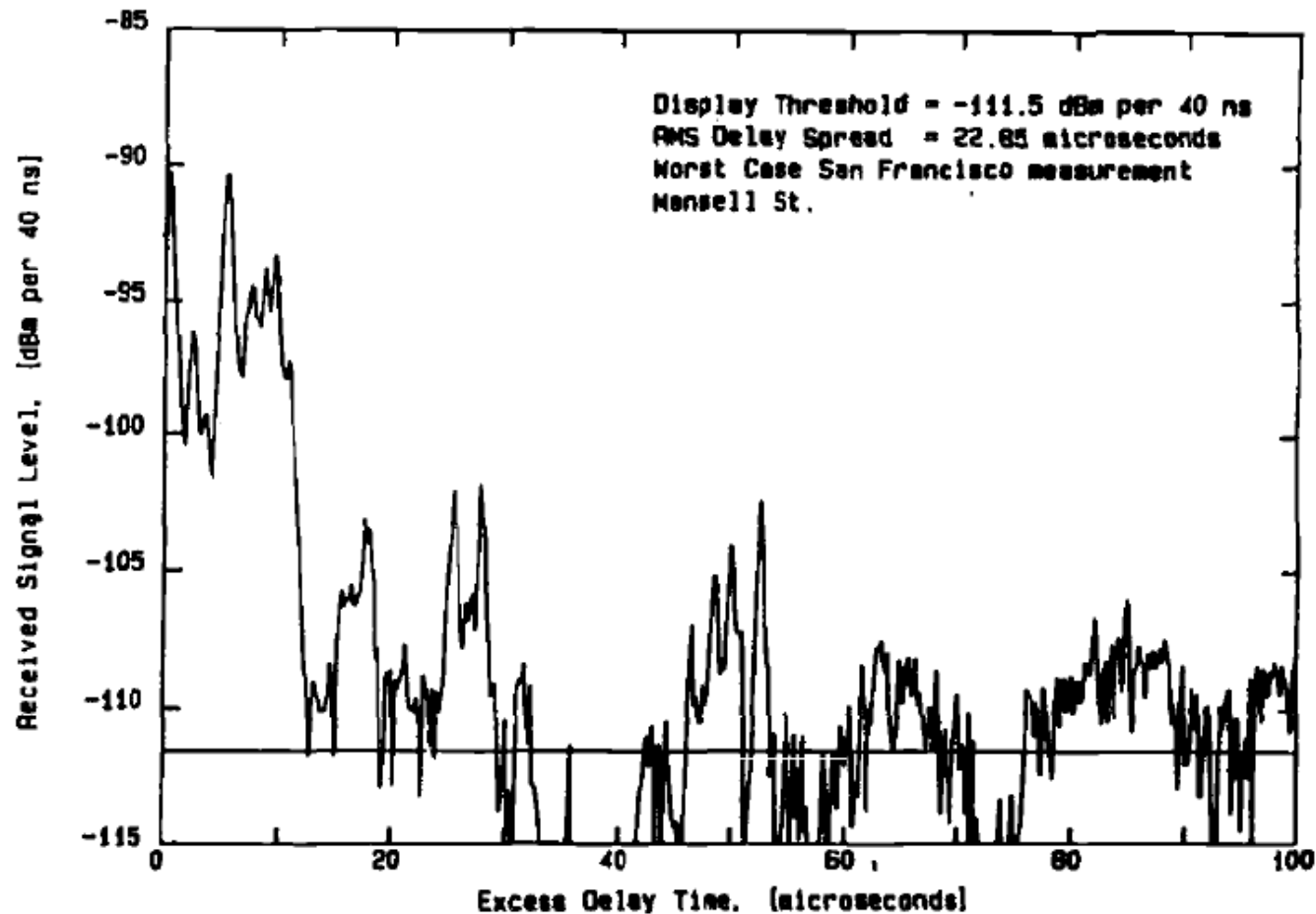
Impulse Response Model of a Multipath Channel

- Self-Study: Relationship Between Bandwidth and Received Power (5.2.1)

Parameters of Mobile Multipath Channels

- Many multipath channel parameters are derived from the power delay profiles, which are represented as plots of relative received power as a function of excess delay with respect to a fixed time delay reference.
- Power delay profiles are found by averaging instantaneous power delay profile measurements over a local area in order to determine an average small-scale power delay profile.

Parameters of Mobile Multipath Channels



Time Dispersion Parameters

In order to compare different multipath channels and to develop some general design guidelines for wireless systems, parameters which grossly quantify the multipath channel are used. The *mean excess delay*, *rms delay spread*, and *excess delay spread* (X dB) are multipath channel parameters that can be determined from a power delay profile. The time dispersive properties of wide band multipath channels are most commonly quantified by their mean excess delay ($\bar{\tau}$) and rms delay spread (σ_{τ}). The mean excess delay is the first moment of the power delay profile and is defined to be

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k}{\sum_k P(\tau_k)} \quad (4.35)$$

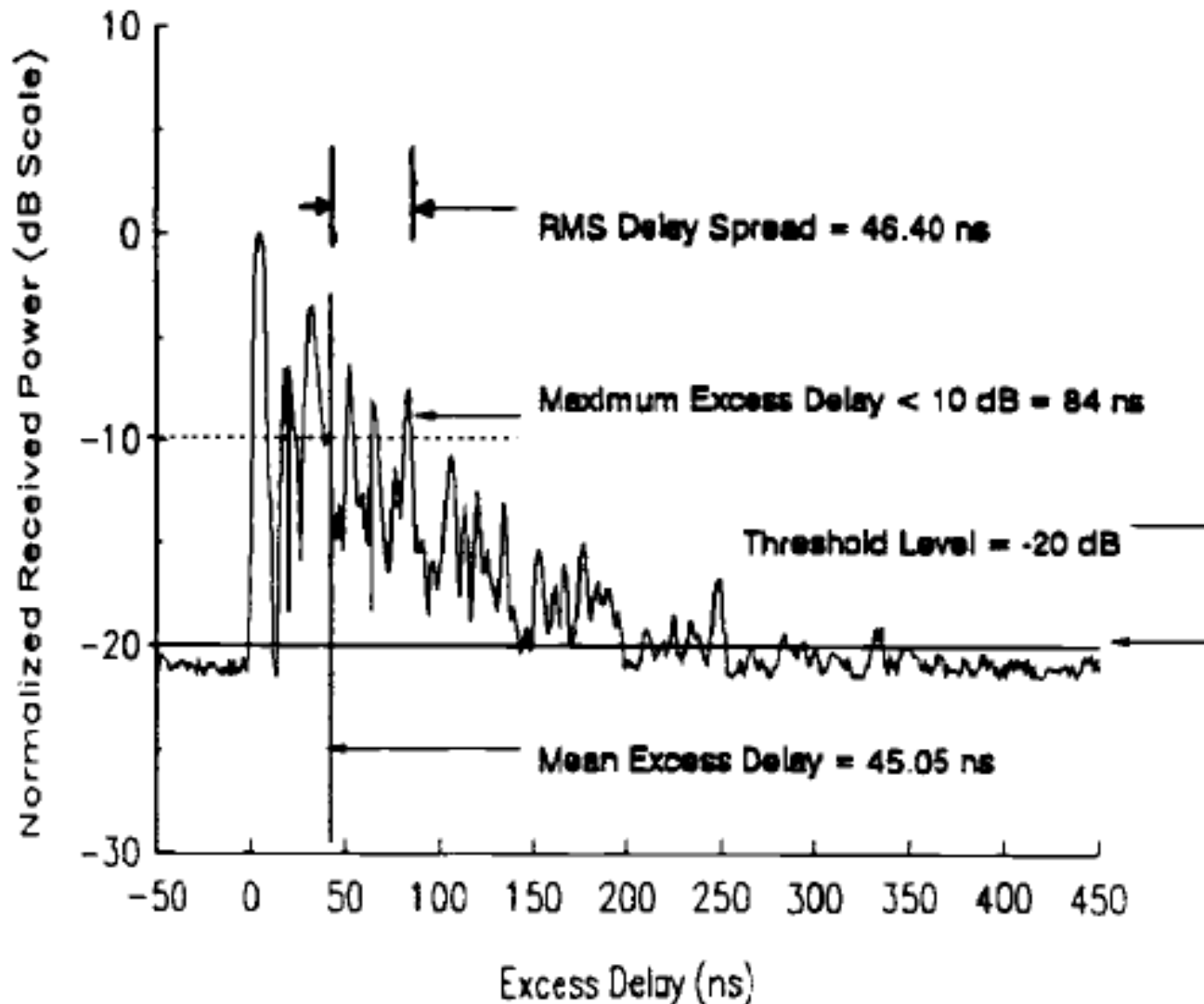
The rms delay spread is the square root of the second central moment of the power delay profile and is defined to be

$$\sigma_{\tau} = \sqrt{\overline{\tau^2} - (\bar{\tau})^2} \quad (4.36)$$

where

$$\overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k^2}{\sum_k P(\tau_k)} \quad (4.37)$$

Time Dispersion Parameters



Time Dispersion Parameters

These delays are measured relative to the first detectable signal arriving at the receiver at $\tau_0 = 0$. Equations (4.35) - (4.37) do not rely on the absolute power level of $P(\tau)$, but only the relative amplitudes of the multipath components within $P(\tau)$. Typical values of rms delay spread are on the order of microseconds in outdoor mobile radio channels and on the order of nanoseconds in indoor radio channels. Table 4.1 shows the typical measured values of rms delay spread.

It is important to note that the rms delay spread and mean excess delay are defined from a single power delay profile which is the temporal or spatial average of consecutive impulse response measurements collected and averaged over a local area. Typically, many measurements are made at many local areas in order to determine a statistical range of multipath channel parameters for a mobile communication system over a large-scale area [Rap90].

Time Dispersion Parameters

In practice, values for $\bar{\tau}$, $\overline{\tau^2}$, and σ_τ depend on the choice of noise threshold used to process $P(\tau)$. The noise threshold is used to differentiate between received multipath components and thermal noise. If the noise threshold is set too low, then noise will be processed as multipath, thus giving rise to values of $\bar{\tau}$, $\overline{\tau^2}$, and σ_τ that are artificially high.

It should be noted that the power delay profile and the magnitude frequency response (the spectral response) of a mobile radio channel are related through the Fourier transform. It is therefore possible to obtain an equivalent description of the channel in the frequency domain using its frequency response characteristics. Analogous to the delay spread parameters in the time domain, *coherence bandwidth* is used to characterize the channel in the frequency domain. The rms delay spread and coherence bandwidth are inversely proportional to one another, although their exact relationship is a function of the exact multipath structure.

Coherence Bandwidth

While the delay spread is a natural phenomenon caused by reflected and scattered propagation paths in the radio channel, the coherence bandwidth, B_c , is a defined relation derived from the rms delay spread. Coherence bandwidth is a statistical measure of the range of frequencies over which the channel can be considered “flat” (i.e., a channel which passes all spectral components with approximately equal gain and linear phase). In other words, coherence bandwidth is the range of frequencies over which two frequency components have a strong potential for amplitude correlation. Two sinusoids with frequency separation greater than B_c are affected quite differently by the channel. If the coherence bandwidth is defined as the bandwidth over which the frequency correlation function is above 0.9, then the coherence bandwidth is approximately [Lee89b]

$$B_c \approx \frac{1}{50\sigma_\tau} \quad (4.38)$$

Coherence Bandwidth

If the definition is relaxed so that the frequency correlation function is above 0.5, then the coherence bandwidth is approximately

$$B_c \approx \frac{1}{5\sigma_\tau} \quad (4.39)$$

It is important to note that an exact relationship between coherence bandwidth and rms delay spread does not exist, and equations (4.38) and (4.39) are “ball park estimates”. In general, spectral analysis techniques and simulation are required to determine the exact impact that time varying multipath has on a particular transmitted signal [Chu87], [Fun93], [Ste94]. For this reason, accurate multipath channel models must be used in the design of specific modems for wireless applications [Rap91a], [Woe94].

Doppler Spread and Coherence Time

Coherence time T_c is the time domain dual of Doppler spread and is used to characterize the time varying nature of the frequency dispersiveness of the channel in the time domain. The Doppler spread and coherence time are inversely proportional to one another. That is,

$$T_c \approx \frac{1}{f_m} \quad (4.40.a)$$

Coherence time is actually a statistical measure of the time duration over which the channel impulse response is essentially invariant, and quantifies the similarity of the channel response at different times. In other words, coherence time is the time duration over which two received signals have a strong potential for amplitude correlation. If the reciprocal bandwidth of the baseband signal is greater than the coherence time of the channel, then the channel will change during the transmission of the baseband message, thus causing distortion at the receiver. If the coherence time is defined as the time over which the time correlation function is above 0.5, then the coherence time is approximately [Ste94]

$$T_c \approx \frac{9}{16\pi f_m} \quad (4.40.b)$$

Doppler Spread and Coherence Time

where f_m is the maximum Doppler shift given by $f_m = v/\lambda$. In practice, (4.40.a) suggests a time duration during which a Rayleigh fading signal may fluctuate

wildly, and (4.40.b) is often too restrictive. A popular rule of thumb for modern digital communications is to define the coherence time as the geometric mean of equations (4.40.a) and (4.40.b). That is,

$$T_C = \sqrt{\frac{9}{16\pi f_m^2}} = \frac{0.423}{f_m} \quad (4.40.c)$$

Types of Small-Scale Fading

- Depending on the relation between the signal parameters (such as bandwidth, symbol period, etc.) and the channel parameters (such as rms delay spread and Doppler spread), different transmitted signals will undergo different types of fading.
- The time dispersion and frequency dispersion mechanisms in a mobile radio channel lead to four possible distinct effects, which are manifested depending on the nature of the transmitted signal, the channel, and the velocity.
- While multipath delay spread leads to time dispersion and frequency selective fading, Doppler spread leads to frequency dispersion and time selective fading. The two propagation mechanisms are independent of one another.

Types of Small-Scale Fading

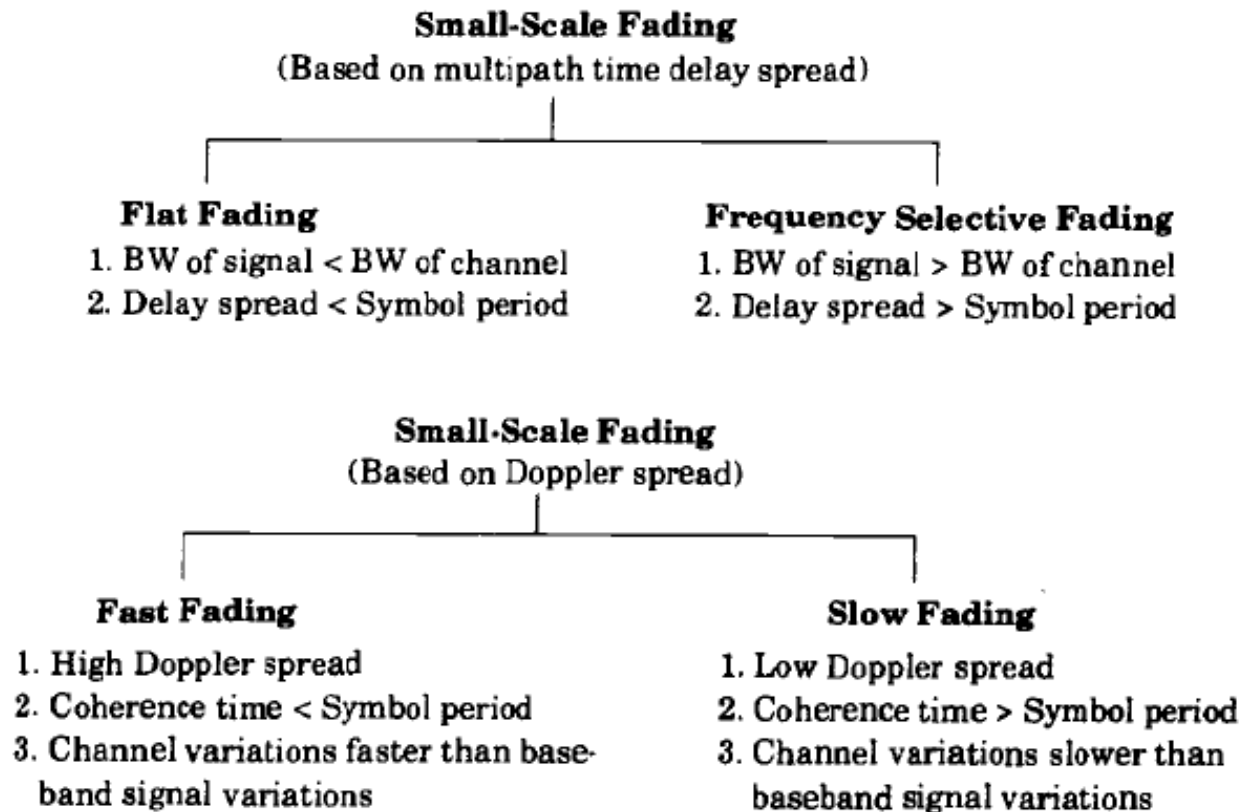


Figure 4.11

Types of small-scale fading.

Fading Effects Due to Multipath time Delay

Spread Flat fading

- If the mobile radio channel has a constant gain and linear phase response over a bandwidth which is greater than the bandwidth of the transmitted signal, then the received signal will undergo flat fading.
- This type of fading is historically the most common type of fading described in the technical literature.
- In flat fading, the multipath structure of the channel is such that the spectral characteristics of the transmitted signal are preserved at the receiver.
- However the strength of the received signal changes with time, due to fluctuations in the gain of the channel caused by multipath.

Flat fading

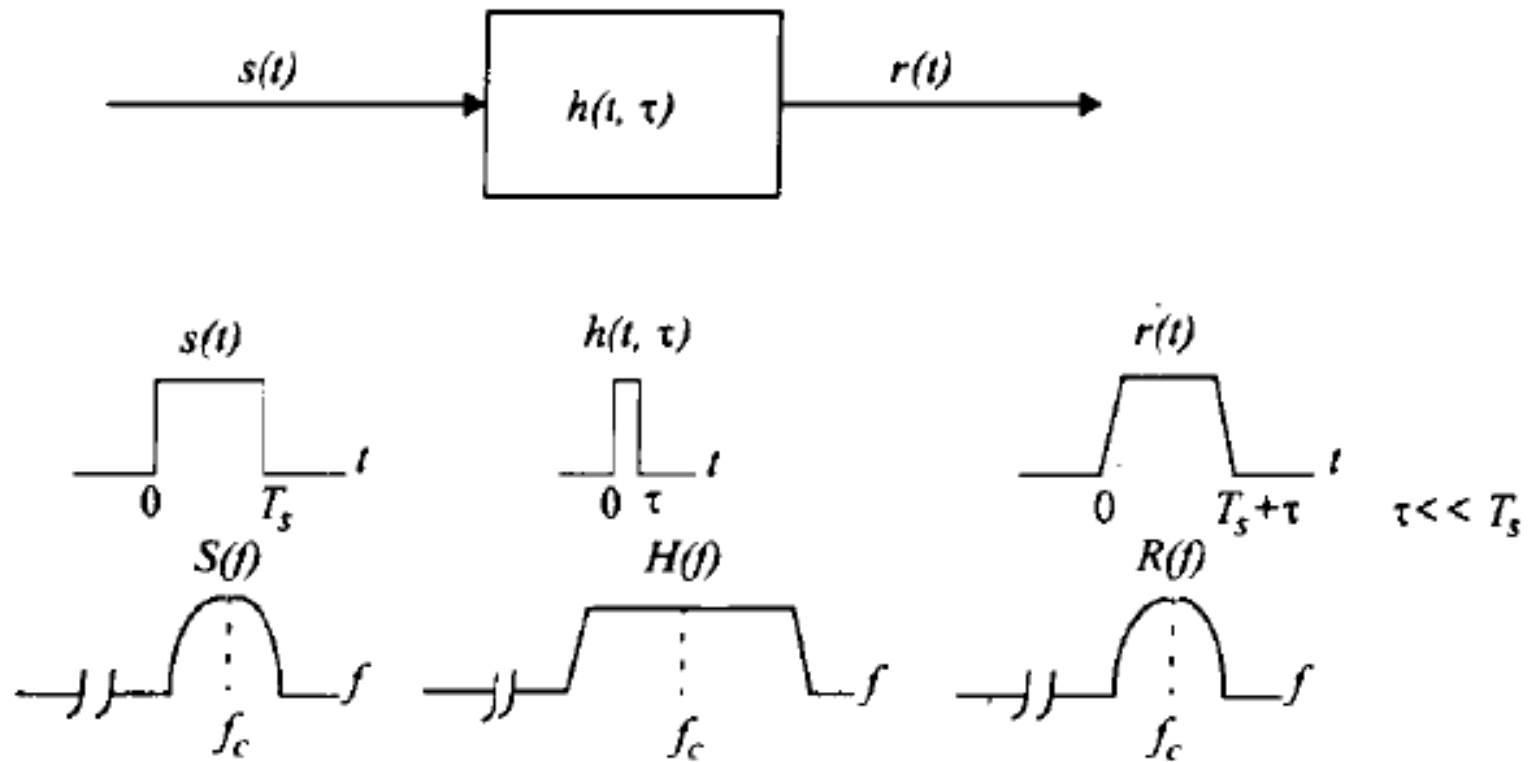


Figure 4.12
Flat fading channel characteristics.

Flat fading

- It can be seen from Figure that if the channel gain changes over time, a change of amplitude occurs in the received signal.
- Over time, the received signal $r(t)$ varies in gain, but the spectrum of the transmission is preserved.
- In a flat fading channel, the reciprocal bandwidth of the transmitted signal is much larger than the multipath time delay spread of the channel, and $h_b(t, \tau)$ can be approximated as having no excess delay (i.e., a single delta function with $\tau = 0$).

Flat fading

- Flat fading channels are also known as amplitude varying channels and are sometimes referred to as narrowband channels, since the bandwidth of the applied signal is narrow as compared to the channel flat fading bandwidth.
- Typical flat fading channels cause deep fades, and thus may require 20 or 30 dB more transmitter power to achieve low bit error rates during times of deep fades as compared to systems operating over non-fading channels.

Flat fading

- The distribution of the instantaneous gain of flat fading channels is important for designing radio links, and the most common amplitude distribution is the Rayleigh distribution.
- The Rayleigh flat fading channel model assumes that the channel induces an amplitude which varies in time according to the Rayleigh distribution.

Flat fading

To summarize, a signal undergoes flat fading if

$$B_S \ll B_C \quad (4.41)$$

and

$$T_S \gg \sigma_\tau \quad (4.42)$$

where T_S is the reciprocal bandwidth (e.g., symbol period) and B_S is the bandwidth, respectively, of the transmitted modulation, and σ_τ and B_C are the rms delay spread and coherence bandwidth, respectively, of the channel.

Frequency Selective Fading

If the channel possesses a constant-gain and linear phase response over a bandwidth that is smaller than the bandwidth of transmitted signal, then the channel creates *frequency selective fading* on the received signal. Under such conditions the channel impulse response has a multipath delay spread which is greater than the reciprocal bandwidth of the transmitted message waveform. When this occurs, the received signal includes multiple versions of the transmitted waveform which are attenuated (faded) and delayed in time, and hence the received signal is distorted. Frequency selective fading is due to time dispersion of the transmitted symbols within the channel. Thus the channel induces *intersymbol interference* (ISI). Viewed in the frequency domain, certain frequency components in the received signal spectrum have greater gains than others.

Frequency Selective Fading

- For frequency selective fading, the spectrum $S(f)$ of the transmitted signal has a bandwidth which is greater than the coherence bandwidth B_c of the channel.
- Viewed in the frequency domain, the channel becomes frequency selective, where the gain is different for different frequency components. Frequency selective fading is caused by multipath delays which approach or exceed the symbol period of the transmitted symbol.

Frequency Selective Fading

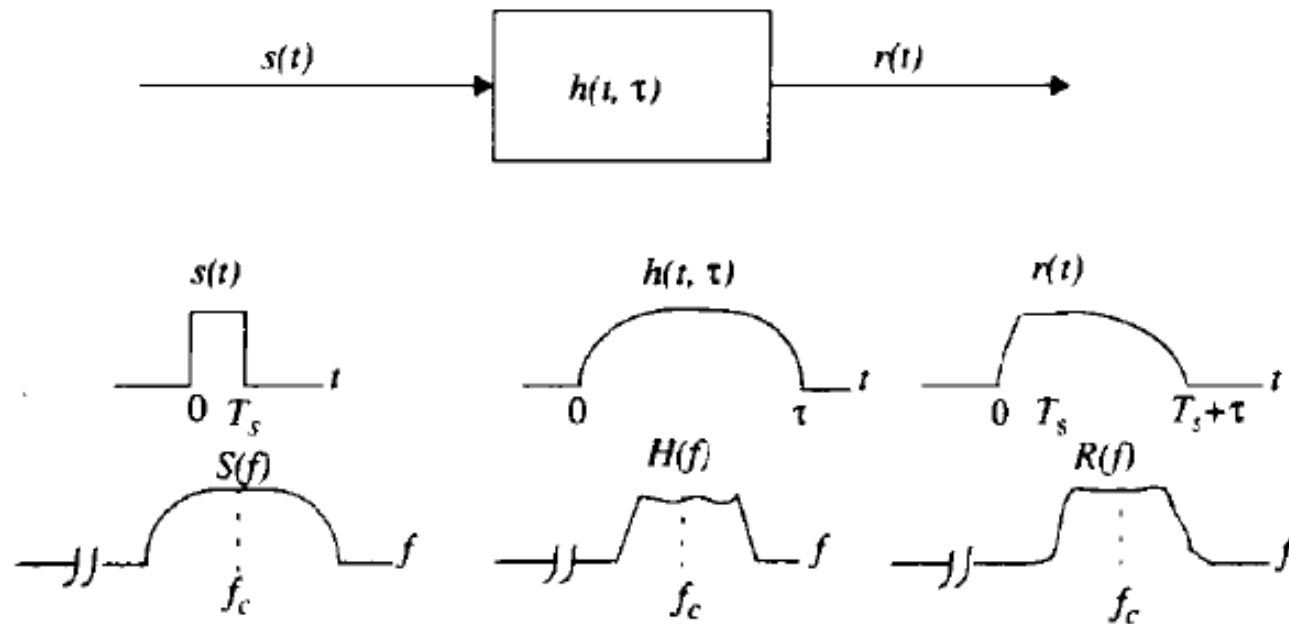


Figure 4.13
Frequency selective fading channel characteristics.

Frequency Selective Fading

period of the transmitted symbol. Frequency selective fading channels are also known as *wideband channels* since the bandwidth of the signal $s(t)$ is wider than the bandwidth of the channel impulse response. As time varies, the channel varies in gain and phase across the spectrum of $s(t)$, resulting in time varying distortion in the received signal $r(t)$. To summarize, a signal undergoes frequency selective fading if

$$B_S > B_C \quad (4.43)$$

and

$$T_S < \sigma_\tau \quad (4.44)$$

A common rule of thumb is that a channel is frequency selective if $T_S \leq 10\sigma_\tau$, although this is dependent on the specific type of modulation used. Chapter 5

Fading Effects Due to Doppler Spread

Fast Fading

- Depending on how rapidly the transmitted baseband signal changes as compared to the rate of change of the channel, a channel may be classified either as a fast fading or slow fading channel.
- In a fast fading channel, the channel impulse response changes rapidly within the symbol duration.
- That is, the coherence time of the channel is smaller than the symbol period of the transmitted signal.
- This causes frequency dispersion (also called time selective fading) due to Doppler spreading, which leads to signal distortion. Viewed in the frequency domain, signal distortion due to fast fading increases with increasing Doppler spread relative to the bandwidth of the transmitted signal.

Fading Effects Due to Doppler Spread

Fast Fading

spread relative to the bandwidth of the transmitted signal. Therefore, a signal undergoes fast fading if

$$T_s > T_c \quad (4.45)$$

and

$$B_s < B_D \quad (4.46)$$

It should be noted that when a channel is specified as a fast or slow fading channel, it does not specify whether the channel is flat fading or frequency selective in nature. Fast fading only deals with the rate of change of the channel due to motion. In the case of the flat fading channel, we can approximate the impulse response to be simply a delta function (no time delay). Hence, a *flat fading, fast fading* channel is a channel in which the amplitude of the delta function varies faster than the rate of change of the transmitted baseband signal. In the case of a *frequency selective, fast fading* channel, the amplitudes, phases, and time delays of any one of the multipath components vary faster than the rate of change of the transmitted signal. In practice, fast fading only occurs for very low data rates.

Fading Effects Due to Doppler Spread

Slow Fading

In a *slow fading channel*, the channel impulse response changes at a rate much slower than the transmitted baseband signal $s(t)$. In this case, the channel may be assumed to be static over one or several reciprocal bandwidth intervals. In the frequency domain, this implies that the Doppler spread of the channel is much less than the bandwidth of the baseband signal. Therefore, a signal undergoes slow fading if

$$T_S \ll T_C \quad (4.47)$$

and

$$B_S \gg B_D \quad (4.48)$$

It should be clear that the velocity of the mobile (or velocity of objects in the channel) and the baseband signaling determines whether a signal undergoes fast fading or slow fading.

The relation between the various multipath parameters and the type of fading experienced by the signal are summarized in Figure 4.14. Over the years, some authors have confused the terms fast and slow fading with the terms large-scale and small-scale fading. It should be emphasized that fast and slow fading deal with the relationship between the time rate of change in the channel and the transmitted signal, and not with propagation path loss models.

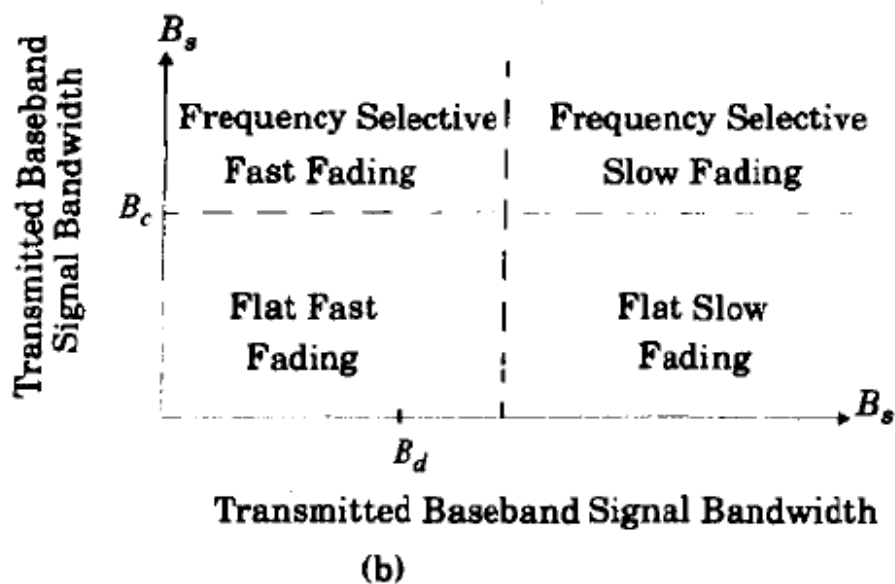
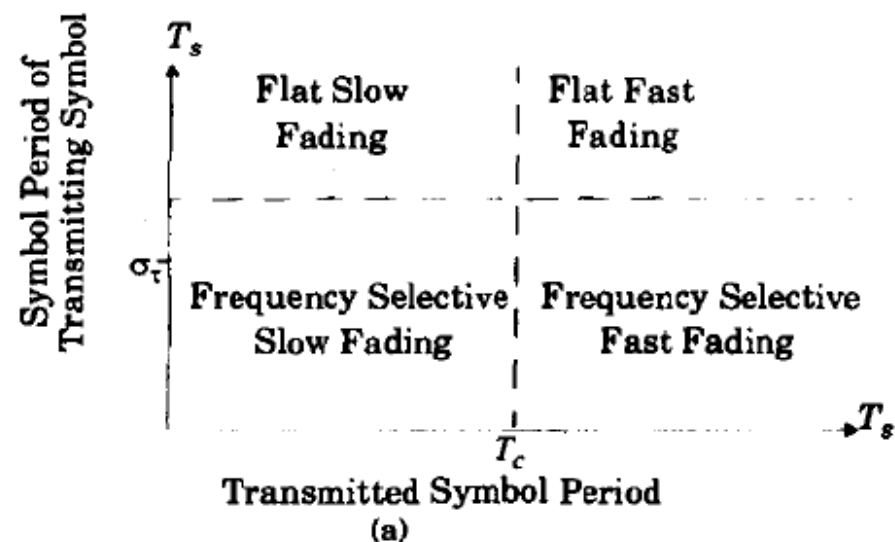


Figure 4.14

Matrix illustrating type of fading experienced by a signal as a function of

(a) symbol period

(b) baseband signal bandwidth.

Rayleigh and Ricean Distributions

- Rayleigh Fading Distribution
- Ricean Fading Distribution

(Self-Study)